# Steve Berggreen

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## Research Interests

Global environmental change; climate impacts; hydrology and water resources; human health and development.
**Methods:** hydrological modeling, geospatial analysis, remote sensing, causal inference, and machine learning.

## Academic Appointments

**Stanford University** January 2025–present

Postdoctoral Scholar, Global Policy Laboratory, Stanford Doerr School of Sustainability
Principal investigator: Solomon Hsiang

## Education

**Lund University** 2024

Ph.D. in Economics
Supervisors: Therese Nilsson and Gunes Gokmen

**University of Gothenburg** 2016

M.Sc. in Economics, with distinction

**Lund University**

M.Sc. in Water Resources Engineering 2011

B.Sc. in Civil Engineering 2009

*Research visit:* London School of Economics, Visiting Research Student, 2023.
*Exchange study:* ENSEEIHT, Hydraulics and Fluid Mechanics, Erasmus scholarship, 2010.

## Professional Experience in Hydrology and Climate Science

**Danish Hydraulic Institute** 2016–2019

Development and application of methods for modeling urban flash floods.

**Swedish Meteorological and Hydrological Institute** 2014–2016

Contributed to applied hydrological and climate-impact modeling for national and regional climate-adaptation assessments, including downscaled climate scenarios, future streamflow, and flood risk.

**WSP Group** 2011–2014

Hydrological consulting and project management in flood risk and water-sensitive planning.

## Peer-Reviewed Publications

**A Robot-Assisted Pipeline to Rapidly Scan 1.7 Million Historical Aerial Photographs**

Sheila Masson, Alan Potts, Allan Williams, **Steve Berggreen**, Kevin McLaren, Sam Martin, Eugenio Noda, Nicklas Nordfors, Nic Ruecroft, Hannah Druckenmiller, Solomon Hsiang, Andreas Madestam, and Anna Tompsett.
*npj Heritage Science* 14, article 123 (2026).

## Job Market Paper

**The Child Health Burden of Unobserved Water**

With Linn Mattisson.

## Working Papers

**Estimating the effect of soil moisture on food production, farmer adaption and famine in Africa**

With Hannah Druckenmiller and Solomon Hsiang.

**When the Right Signal Fails: Climate Risk, Precautionary Saving, and the Limits of Self-Insurance**

Under review.

**The Agricultural Origins of Time Preference: Comment**

Revise and resubmit, American Economic Review.

**Demand for Slaves, Markets, and Social Betrayal: How the Slave Trade Shaped Mistrust in Africa**

With Gunes Gokmen.

Under review.

**Traditional Norms and Parental Investment in Human Capital**

With Gunes Gokmen.

## Research in Progress

**Climate Adaptation in Africa over the 20th Century**

With Hannah Druckenmiller, Dylan Hogan, and Solomon Hsiang.

Lead author, Aerial History Project.

**Surface Water and Malaria Prediction in Uganda**

With Shakira Babirye, Gabby Barratt Heitmann, Jade Benjamin-Chung, Grant Dorsey, Paul Krezanoski, Henry Mawejje, Erin Mordecai, Joaniter Nankabirwa, Mercy Opiyo, and Eddie Thomsen.

Field data collection ongoing.

**The Gravity of Water: Water Scarcity and the Economic Geography of Floodplains**

Manuscript in preparation.

**One Million High-Resolution Aerial Photographs Capture Africa's 20th-Century Transformation**

With Eugenio Noda, Hannah Druckenmiller, Joel Ferguson, Simon Greenhill, Solomon Hsiang, Taylor Kee, Andreas Madestam, Hikari Murayama, Nicklas Nordfors, Stephan Schneider, and Anna Tompsett.

Manuscript in preparation, Aerial History Project.

**Cuisine**

With Girija Borker and Margarita Gafaro.

Manuscript in preparation.

## Policy Publications and Proceedings

**Stagnant Water and Waterborne Diseases: Implications for Children's Health and Education**

EBA Expert Group for Aid Studies, Development Dissertation Brief, 2025.

**Traditionella Normer och Föräldrars Investering i Humankapital**

With Jan Bietenbeck and Gunes Gokmen.

*Ekonomisk Debatt*, 2023; papers and proceedings of the 2022 Swedish National Conference in Economics.

## Teaching and Mentoring

**Research mentoring and supervision**

**2026–** Undergraduate research mentor, Stanford University (ongoing).
**2026** Undergraduate research mentor, Stanford SURGE (Khussam Mohamed).
**2019** M.Sc. thesis supervisor for Lisa Olsson, Water Resources Engineering.

**Teaching assistantships**

**2020–2024** Financial Economics (multiple course offerings).
**2021–2023** Foundations of Finance.
**2020–2022** Microeconomic Theory with Applications.
**2019–2020** Macroeconomic Theory and Economic Policy.
**2011** Water and Wastewater Technology.

## Fellowships, Grants, and Awards

**2026** Stanford SURGE program funding, USD 5,000, supporting undergraduate research at Stanford.

**2023** Conference grant, Stiftelsen för främjande av ekonomisk forskning, SEK 7,500.

**2022** Hedelius Scholarship for postgraduate studies at foreign universities, SEK 300,000.

**2022** Paper selected for the Swedish National Conference in Economics papers and proceedings, *Ekonomisk Debatt*.

**2016** Malmsten Award for Best M.Sc. Thesis in Economics, SEK 10,000. Thesis: *Flood risk and market property valuation when insurance is free*.

**2009** Erasmus scholarship for studies at ENSEEIHT, Toulouse, SEK 40,000.

## Invited Talks

**2024** 1st Development Economics Workshop, Durham University Business School.

**2023** Development Economics Research Group, University of Copenhagen; IFS/STICERD (LSE)/UCL Development Work-in-Progress Seminar, IFS; Economics of the Energy and Environment Seminar, LSE; Copenhagen Business School; ENS Lyon.

**2022** Young Leaders Programme Personal Finance Workshop, Friedrich Naumann Foundation.

## Conference and Seminar Presentations

**2026** NEUDC, Cornell University (upcoming); 6th Arne Ryde Workshop on Culture, Institutions and Development, Lund (upcoming); Midwest International Economic Development Conference, University of Wisconsin–Madison; Sustainability Data Science Conference, Stanford; GradSEEDS Seminar, Stanford.

**2025** Environmental and Energy Economics Festival, Stanford; WASH Econ, Oberlin College; Global Environmental Policy Student Seminar, Stanford.

**2024** 8th Swedish Conference in Economics, Lund; PhD Conference on Sustainable Development, Lund; Development Tea, Stockholm University; Brown Bag Seminar, Lund (twice).

**2023** Southern Economic Association Annual Meeting, New Orleans; CESifo Economics of Children Workshop, LMU Munich; NEUDC, Harvard; EALE, Prague; EEA, Barcelona; CSAE, Oxford; 3rd Arne Ryde Workshop on Culture, Institutions and Development, Lund; Essen Health Conference, University of Duisburg–Essen; Nordic Conference in Development, Gothenburg; EUDN PhD Workshop, Stockholm; DEM Workshop on Ageing, Culture and Comparative Development, University of Luxembourg; Economic Geography PhD Seminar, LSE; SWELL Seminar Series in Water Economics; Copenhagen Education Network Workshop, Lund.

**2022** Swedish Conference in Economics, Stockholm School of Economics; IAST Workshop in Political Economy, Toulouse School of Economics; ASWEDE 7th Annual Conference, Stockholm; 2nd Arne Ryde Workshop on Culture, Institutions and Development, Lund; Brown Bag Seminar, Lund (twice); Toulouse Summer

School in Quantitative Social Sciences, Toulouse School of Economics; ASREC Graduate Student Workshop in the Economics of Religion, IRES, Chapman University.

**2021** Arne Ryde Workshop on Culture, Institutions and Development, Lund; Copenhagen Education Network Workshop, Lund; Scandinavian PhD Workshop in Economics of Education.

## Academic Service

**Journal refereeing:** Journal of Environmental Economics and Management; Economics & Human Biology.

**2023–2024** Member, EDR doctoral student union election committee.
**2023** Co-organizer, 3rd Arne Ryde Workshop on Culture, Institutions and Development; co-organizer, Economics PhD lunch seminar series, Lund University.
**2022–2023** EDR doctoral student representative, Faculty Research Committee.
**2022** Co-organizer, 2nd Arne Ryde Workshop on Culture, Institutions and Development; PhD representative, Head of Department election committee.
**2021–2022** PhD representative, Department Board.
**2019** Co-organizer, ASREC Europe Conference.

## Technical Skills

Python, Stata, R; Google Earth Engine; ArcGIS, QGIS; LaTeX.

## Languages

Swedish (fluent), English (fluent), French (advanced), Farsi (intermediate).

## Academic References

**Solomon Hsiang**

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**Hannah Druckenmiller**

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**Gunes Gokmen**

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